

# Agenda

virtualisation  
tools

tech-  
needed  
(adv)

Disadvantages  
Problems

Solution

containerisation

Adv  
&  
Disadv

tools

Rev

feature-branch

PR  
↓  
pull  
Request

Main-branch

CI

File

script  
test

git  
push

DEV A  
DEV B

local  
machine

- Build (compile)
- basic tests

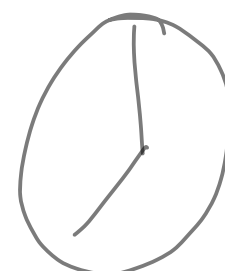
SCM

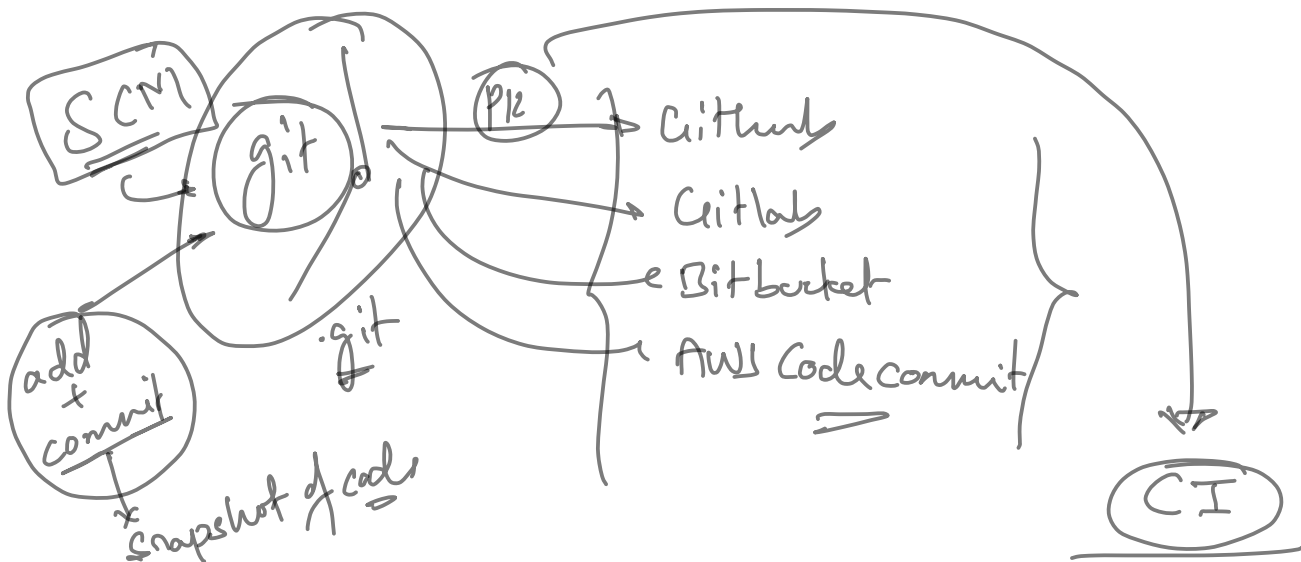
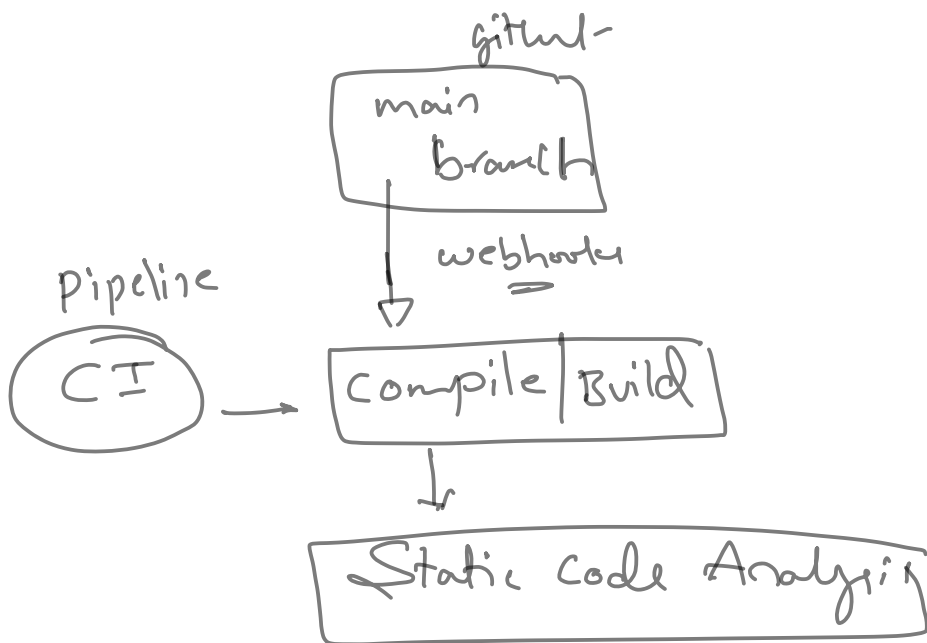
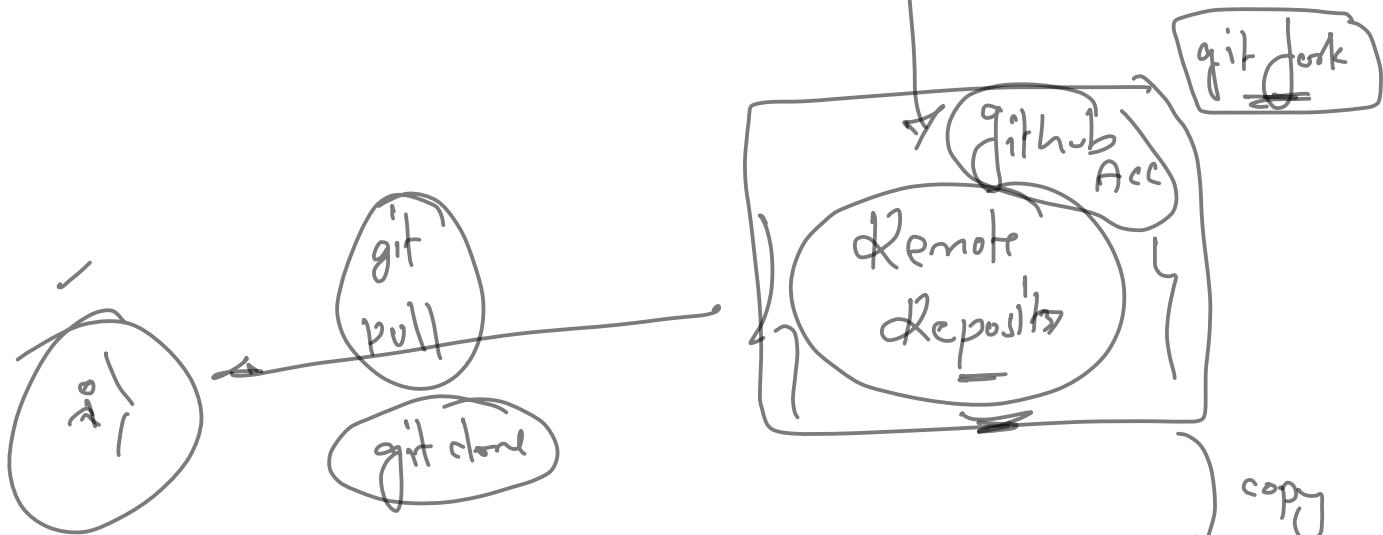
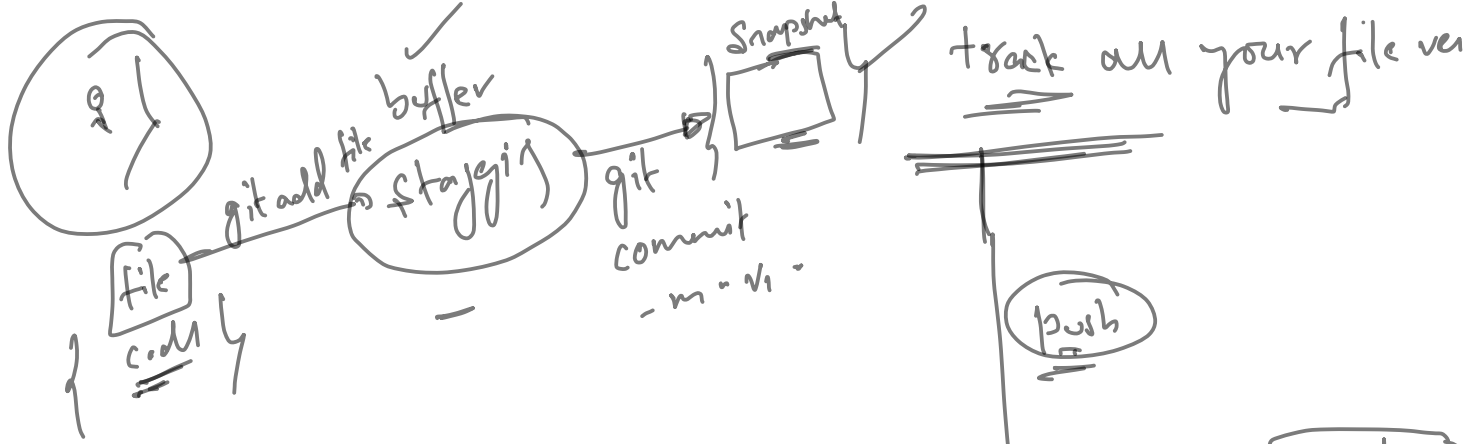
git & github

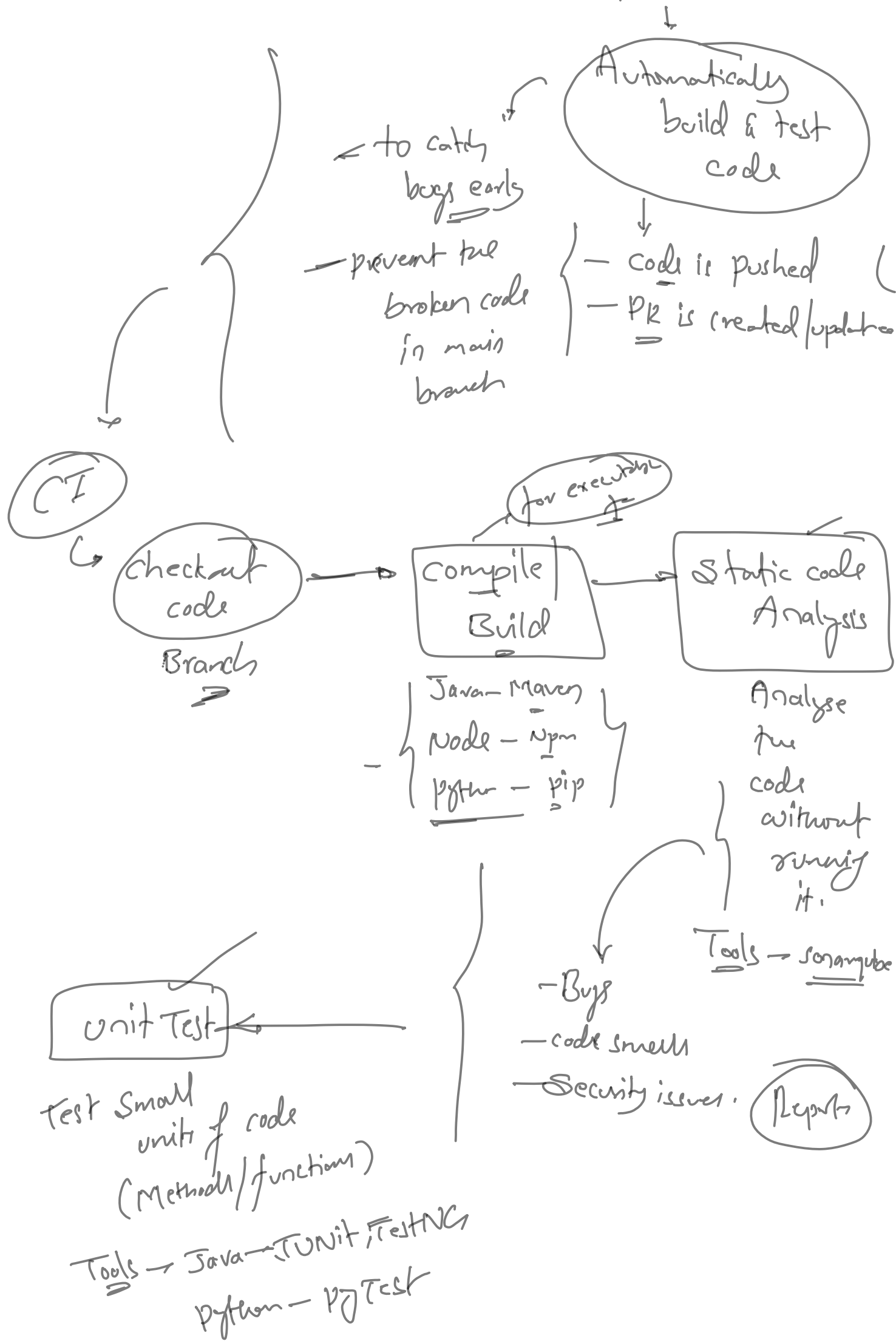
git init

.git

local repository





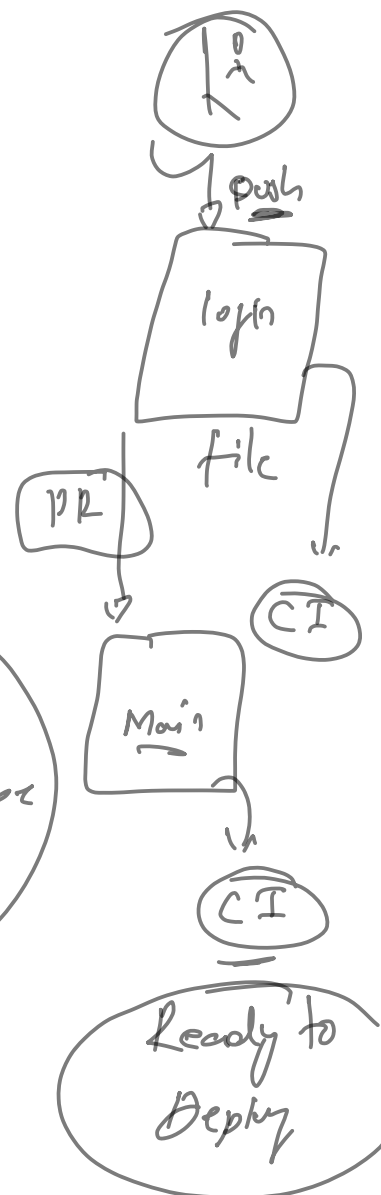
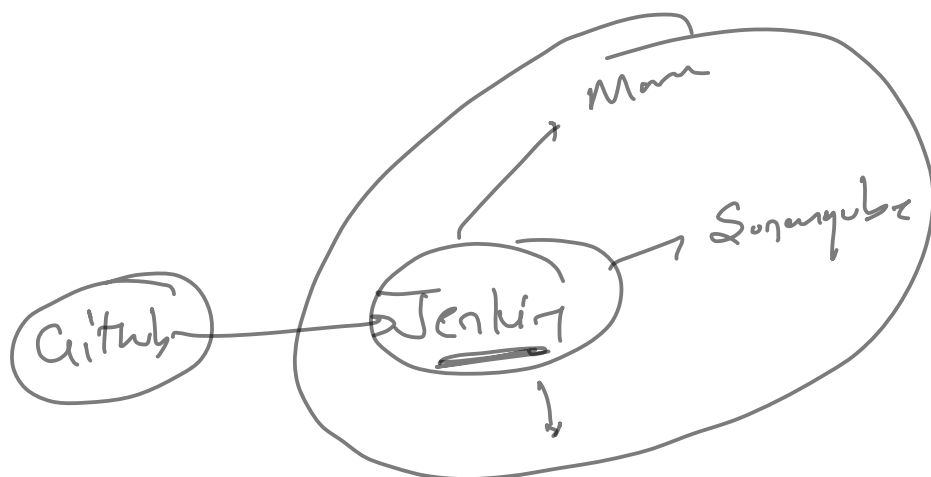


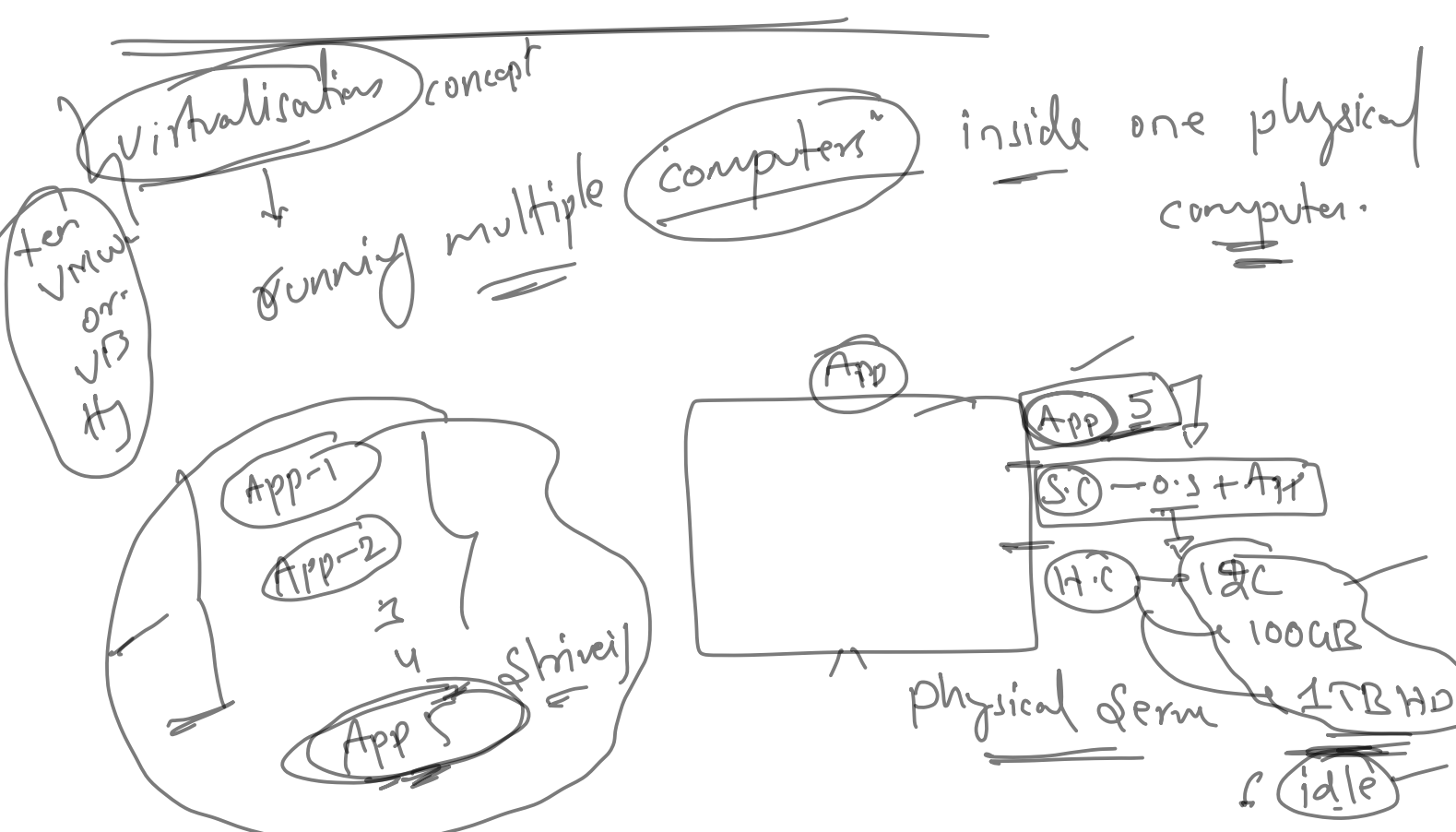
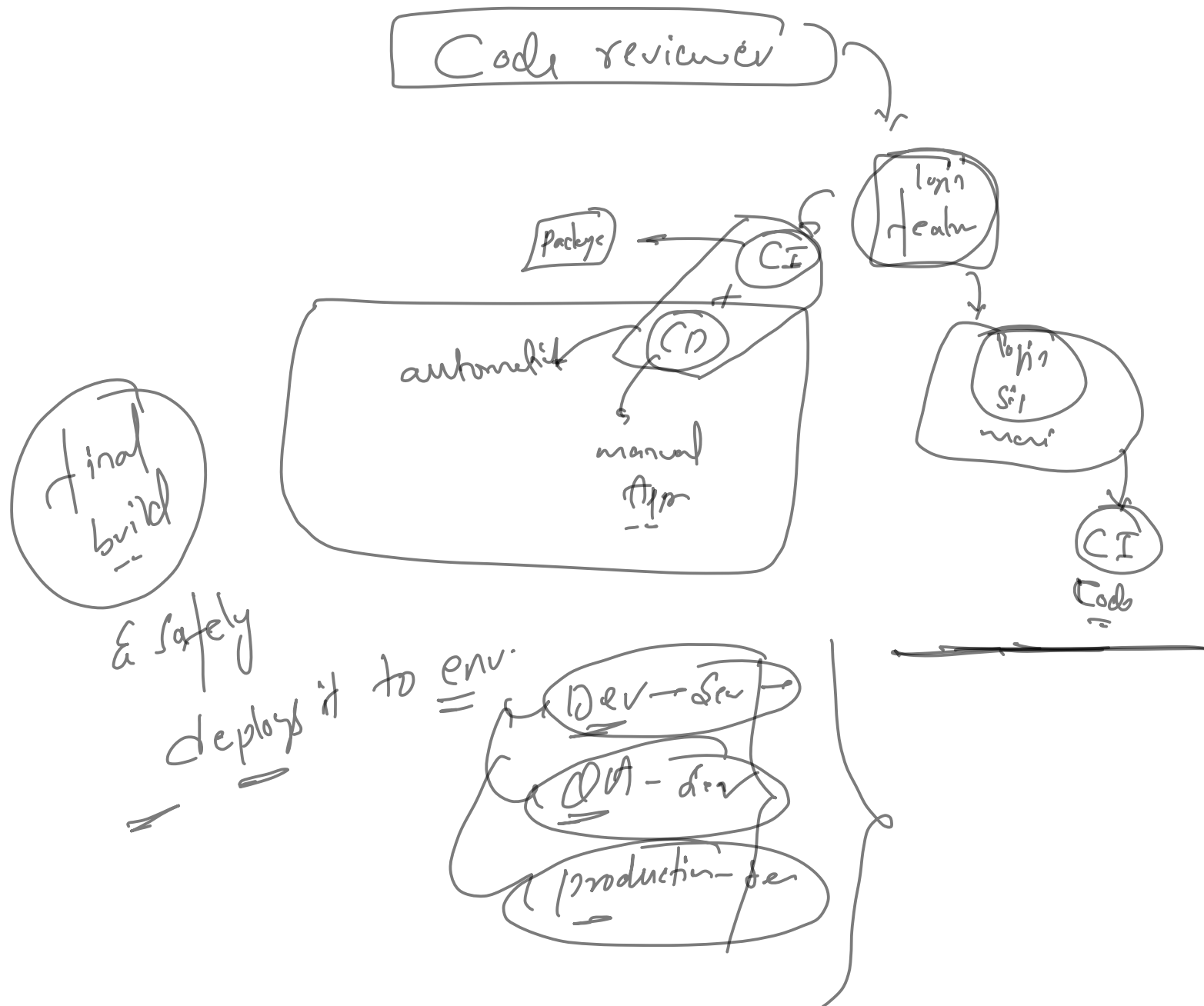
↓  
Code Coverage It Measures how much code is tested

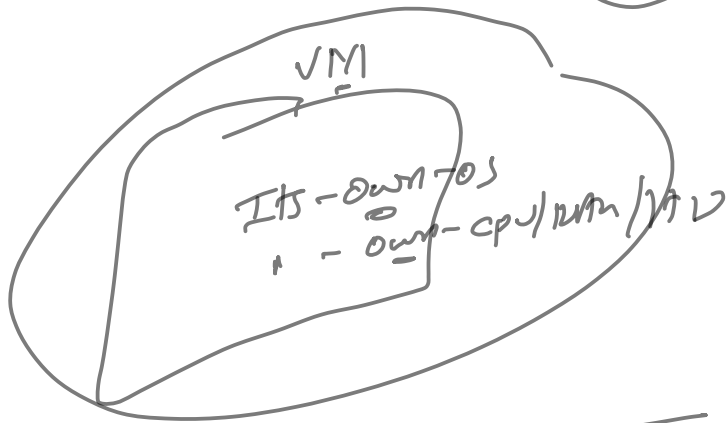
Tools - JacoCo for Java  
Istanbul (JS)  
Coverage.py (python)

CI tool (orchestration)

Jenkins  
Github Actions  
AWS CodeBuild  
Github CI



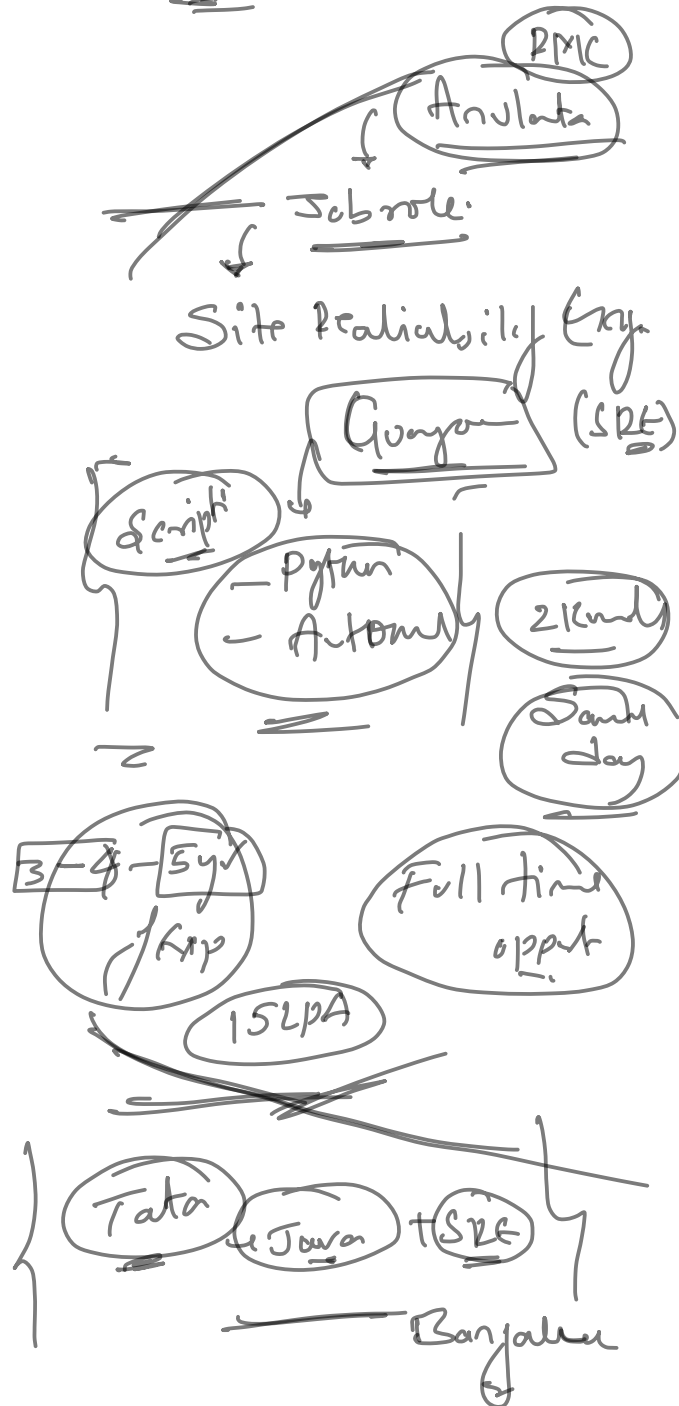




## Advantages

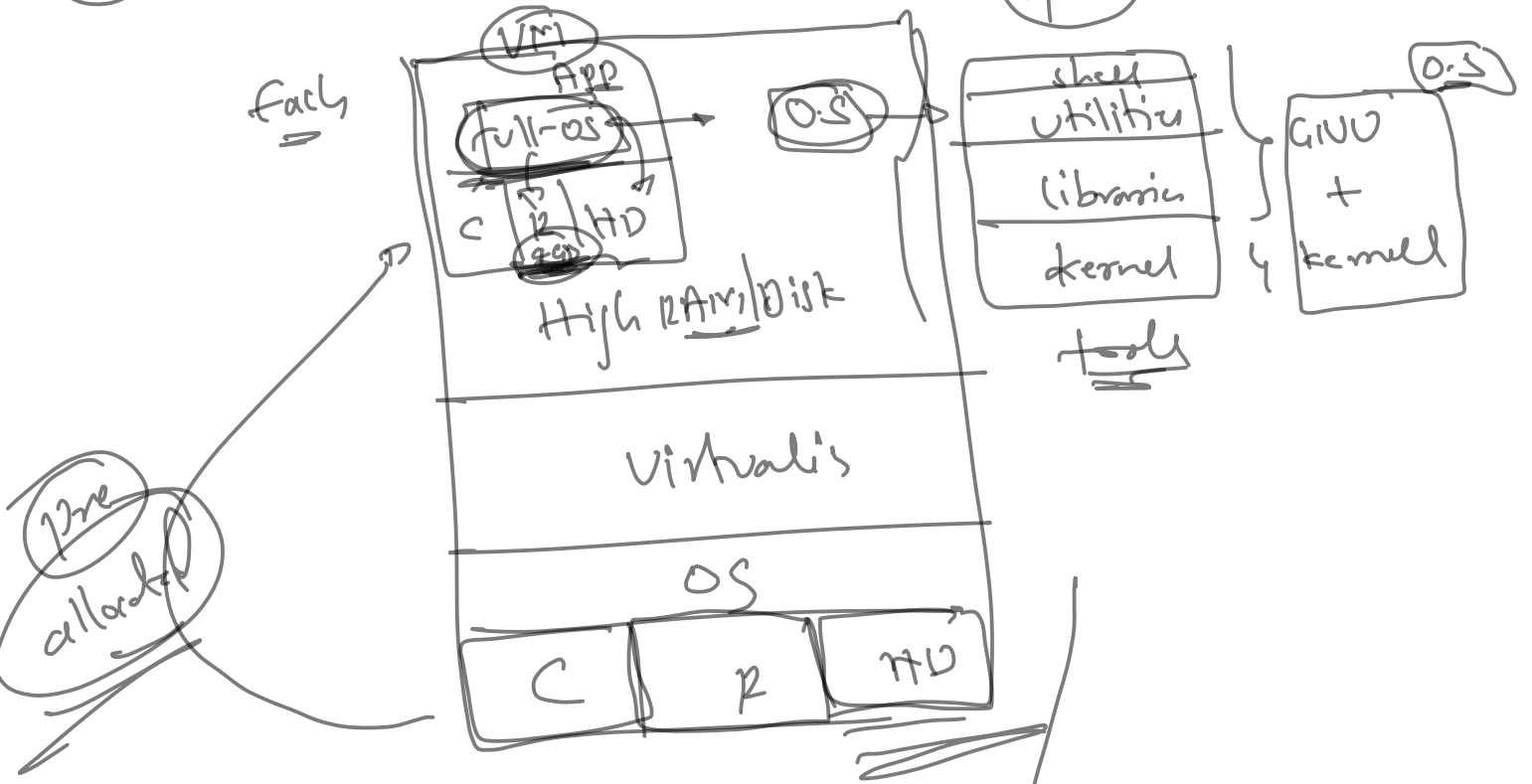
- Better hardware utilization
- Isolation b/w Applications

## Disadvantages of virtualization



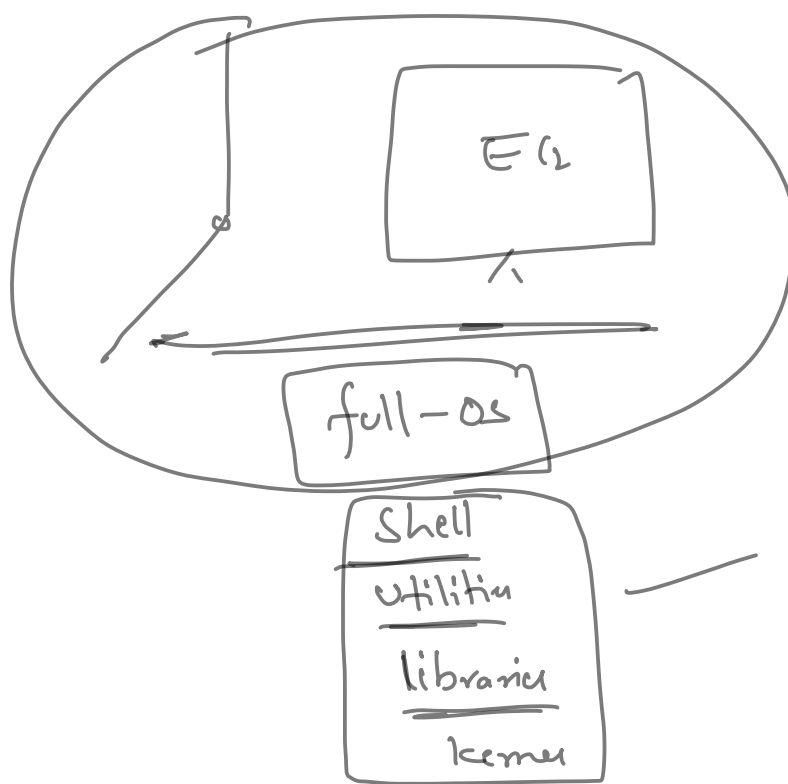
this is where containers come into the picture

pt1 - Heavy OS duplicat.

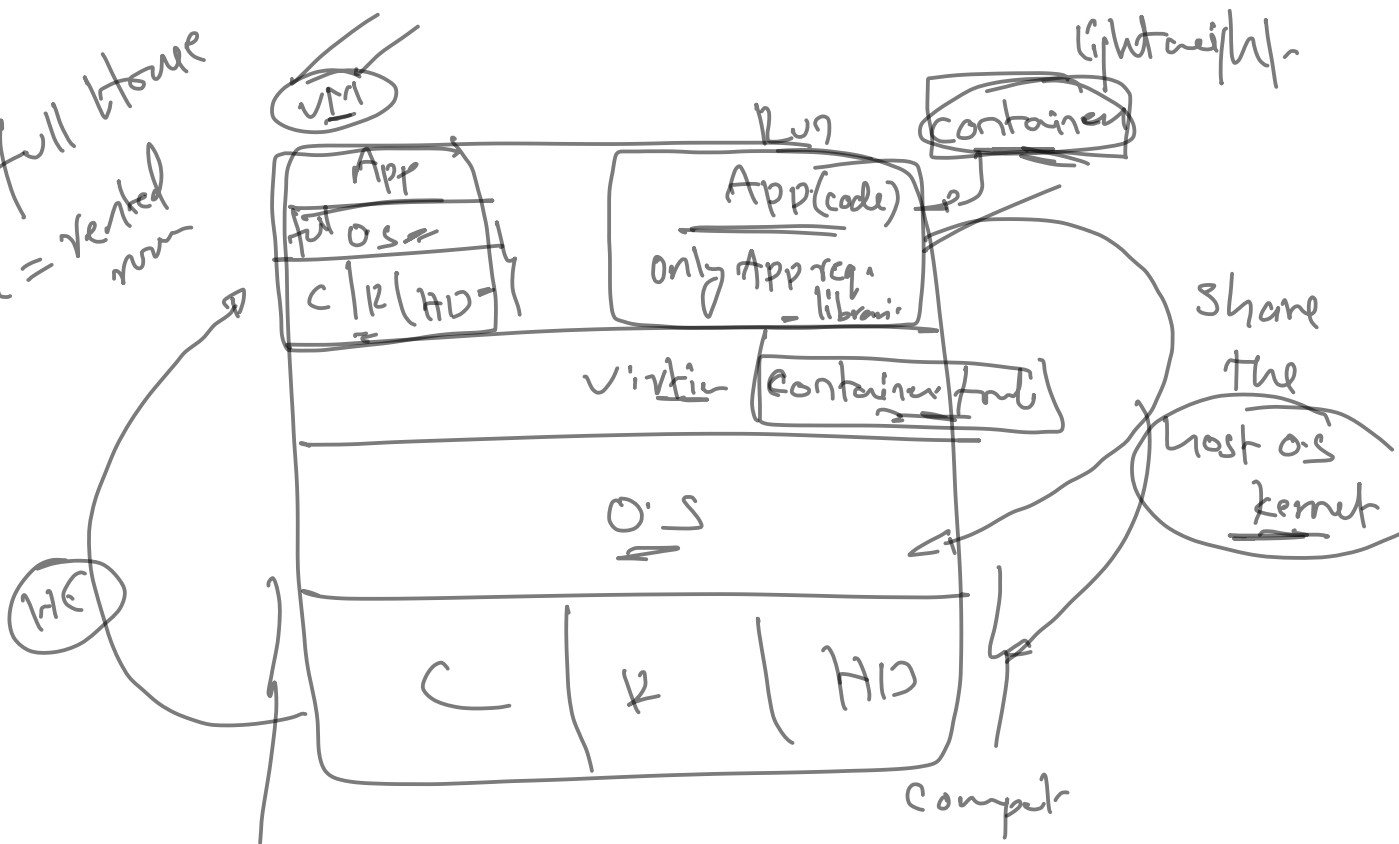


pt2 - Resource wastage

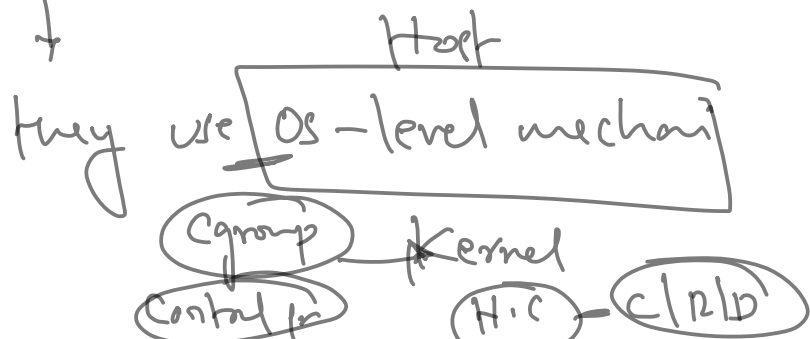
Containerisation  
↓  
running the application in isolated environment without a full os.



VM = full house  
 container = rented room



Containers don't get fixed hardware like VM.





All containers use same physical  
Disk.

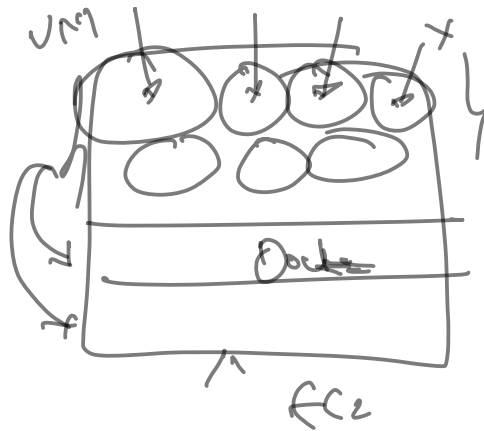
filesystem namespace

VM  
waste RL

VM → 8GB  
App → 1GB  
7GB - Idle

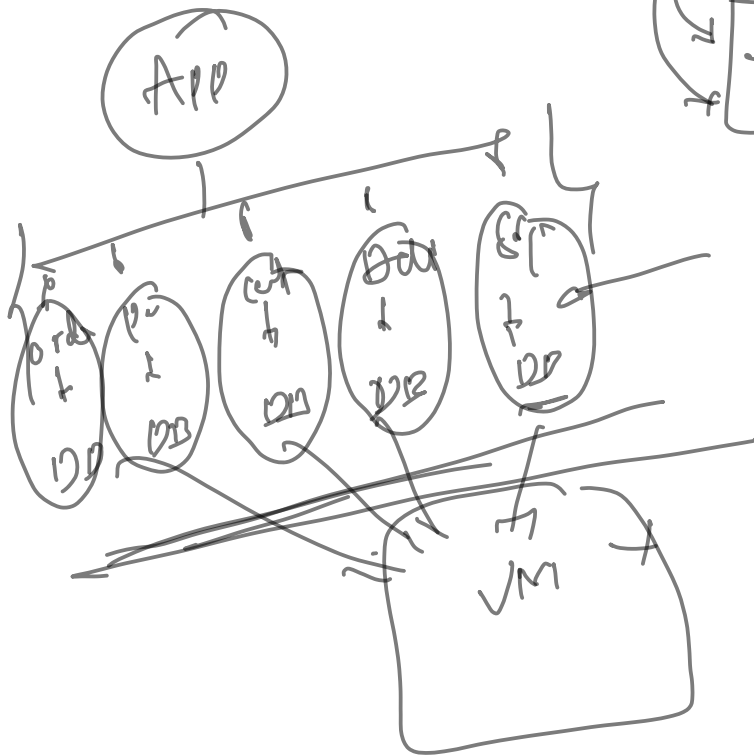
vs

containers use only what it needs

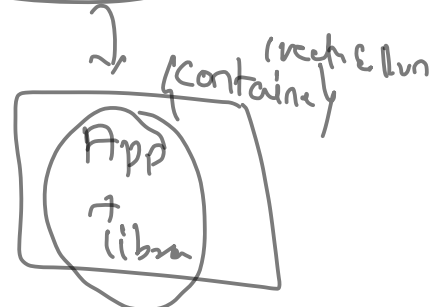


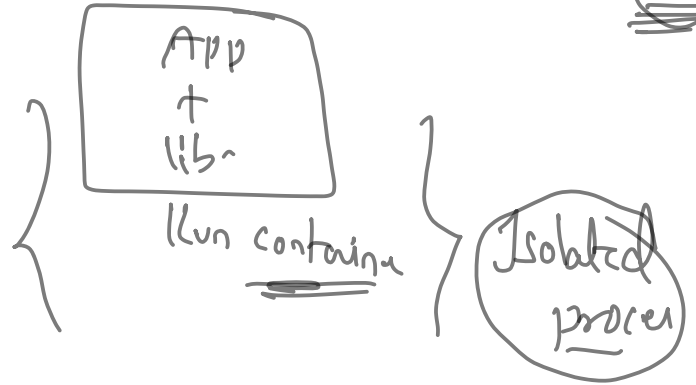
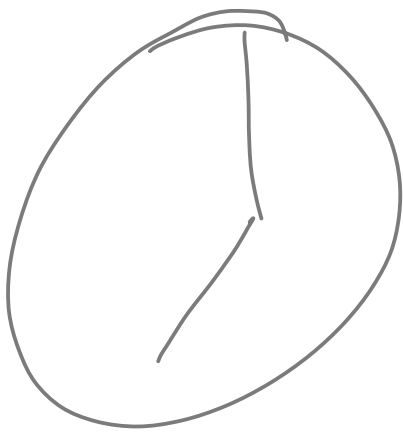
Containers use only  
what it needs

Docker  
podman  
Containerd  
CRI-O



Docker



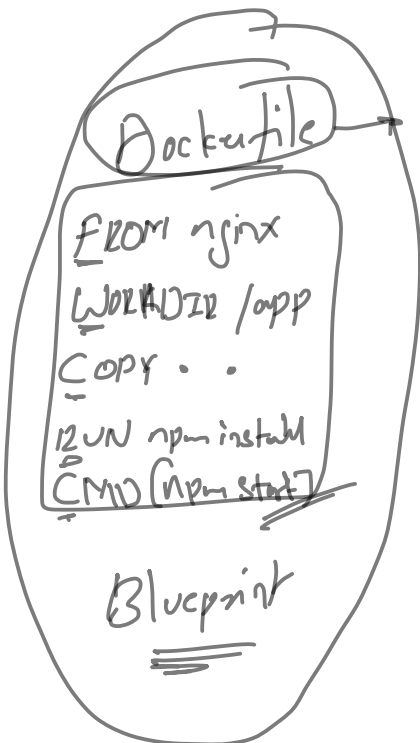


Docker file

Build

Docker Image

re-usable  
template



It is a text file that contains step-by

Step  
Instruction

to  
create a

containers  
Image

Docker build



Docker-Image